

$$S_4/V_4 \simeq S_3,$$

$$V_{\text{pair}}^{(2)} = \text{PROVEN},$$

$$C_3 \hookrightarrow S_3 = \text{PROVEN},$$

$$\text{2-lift on the 17 relations} = \text{PROVEN},$$

$$\text{Squier 3-subcomplex} = \text{PROVEN},$$

$$\text{Peiffer on that 3-subcomplex} = \text{PROVEN}.$$